

Tangent Space.

Def. Given a point $p \in \mathbb{R}^3$ and a vector $V \in \mathbb{R}^3$, a tuple

$V_p = (p, V)$ is called a tangent vector at p to V .

And, the set $T_p \mathbb{R}^3 \stackrel{\text{def}}{=} \{V_p \mid V \in \mathbb{R}^3\}$ is called a tangent space of $\mathbb{R}^3$ at p .

Note: the $T_p \mathbb{R}^3$ is isomorphic to $\mathbb{R}^3$, for each p .

Def. A map $V: \mathbb{R}^3 \rightarrow \bigsqcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3; p \mapsto V_p$ is called a vector field.

Note: for fixed $p \in \mathbb{R}^3$, the space $T_p \mathbb{R}^3$ is generated by a triple:

$$\{e_1(p), e_2(p), e_3(p)\}$$

where $e_1(p) = (1, 0, 0)_p$, $e_2(p) = (0, 1, 0)_p$, $e_3(p) = (0, 0, 1)_p$

Def. Given a differentiable map $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ and the tangent vector V_p ,

$$(f'(p; V)) = V_p[f] = \lim_{t \rightarrow 0} \frac{f(p + tV) - f(p)}{t}$$

is called the derivative of f with respect to V_p .

Lemma. If $V_p = (V_1, V_2, V_3)_p$ is a tangent vector and $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ is differentiable,

then $V_p[f] = \sum_{i=1}^3 V_i \cdot \frac{\partial f}{\partial x_i}(p)$.

Proof. $V_p[f] = f'(p; V) = \underbrace{\begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \frac{\partial f}{\partial x_3} \end{bmatrix}}_{Df(p)} \cdot \underbrace{\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}}_V$

